SL3535R830-1-3D-1WY

3.5x3.5mm, Infrared LED

Ceramic Package Top View LED

SEAM Semiconductors

Technical Data Sheet

Features

- Small SMT ceramic package with high efficiency.
- Low thermal resistance as low as 6℃/W.
- Soldering method: SMT.
- Binning Parameters: Brightness, Forward.
- Voltage, Wavelength and Chromaticity.
- Moisture Sensitivity Level
- Matches ANSI binning.
- Reflow soldering with JEDEC JSTD-020C compatible.
- The product itself will remain within RoHS compliant Version.

Descriptions

The SL3535 series is a surface-mount high-power device featuring high brightness combined with a compact sizethat is suitable for all kinds of lighting applications such as general illumination, flash, spot, signal, industrial andcommercial lighting. The thermal pad of this device is electrically isolated providing convenience in thermal andelectrical design.The SL3535 series is one of the most promising devices in Seam high power product offering and is ready toface the challenges of today's Solid-State Lighting requirements.

Applications

- General Lighting.
- Smart home
- Iris/face recognition
- Biomedical
- Infrared communication
- Infrared monitor, infrared therapy

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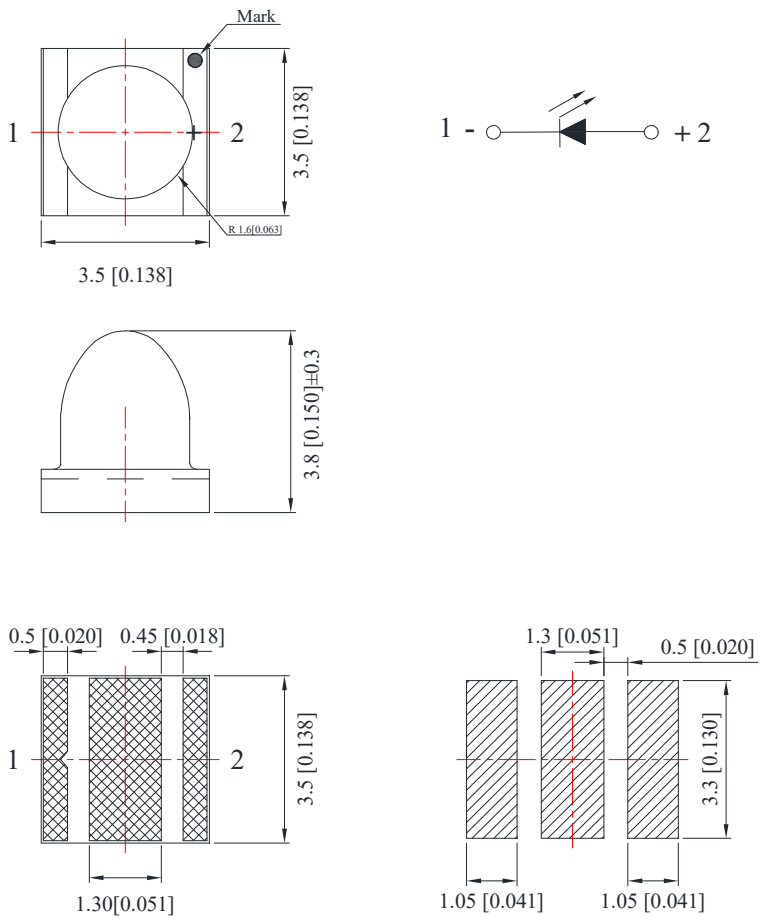
Part No.	Emitting Color	Lens Color
SL3535R830-1-3D-1WY	Infrared	Water Clear

Package Dimension



Notes:

1. All dimensions are in millimeters (inches).
2. Tolerance is ± 0.25 mm (.010″) unless otherwise noted.

Absolute Maximum Ratings at Ta=25℃

Parameters	Symbol	Max	Unit
Power Dissipation	Pd	1	W
Peak Forward Current[(a)]	IFP	500	mA
DC Forward Current[(b)]	IF	350	mA
Reverse Voltage	VR	5	V
Electrostatic Discharge (HBM)	ESD	2000	V
LED Junction Temperature	Tj	120	℃
Operating Temperature Range	Topr	-40℃ to +85℃	
Storage Temperature Range	Tstg	-40℃ to +100℃	
Soldering Temperature	Tsld	260℃ for 5 Seconds	

Notes:

a. Duty Factor = 10%, Frequency = 1 kHz

b. Derate linearly as shown in derating curve.

Electrical Optical Characteristics at Ta=25℃

Parameters	Symbol	Min.	Typ.	Max.	Unit	Test Condition
Radiant Flux[(a)]	Φv	225		275	mW	IF=350mA
Viewing Angle	2θ1/2	---	30	---	Deg	IF=350mA
Peak Emission Wavelength[(b)]	λp	828	830	832	nm	IF=350mA
Spectral Line Half-Width	△λ	---	15	---	nm	IF=350mA
Forward Voltage[(C)]	VF	1.60		2.00	V	IF=350mA
Thermal Resistance	Rth j-s	---	6.0	---	℃/W	IF=350mA
Reverse Current	IR	---	---	50	μA	VR=5V

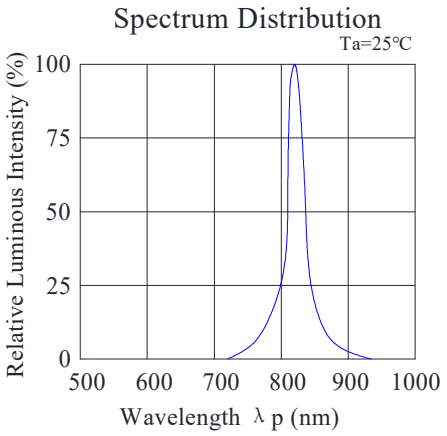
Notes:

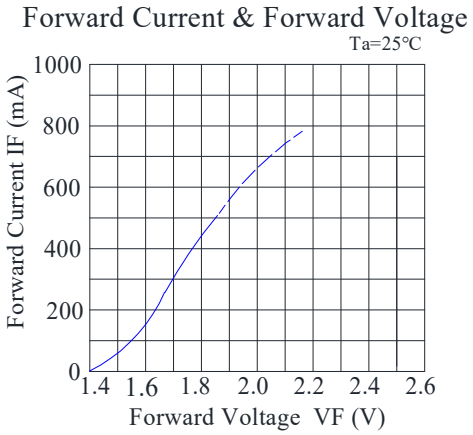
a. Radiant Flux measurement tolerance: ±10%.

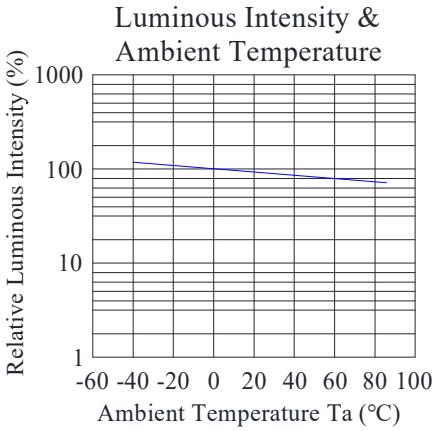
b. Wavelength measurement tolerance: ±1nm

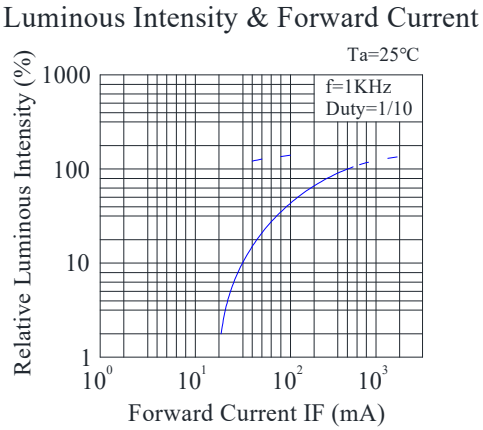
c. Forward voltage measurement tolerance: ±0.1V

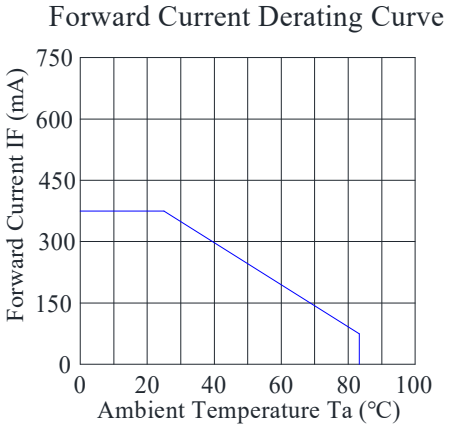
Typical Electrical / Optical Characteristics Curves
(25℃ Ambient Temperature Unless Otherwise Noted)

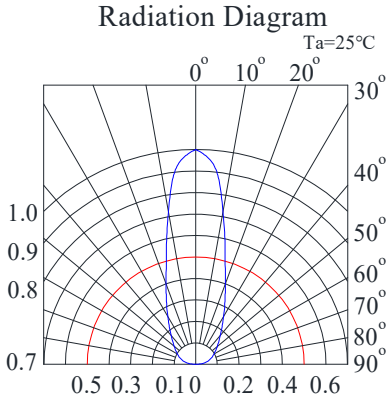




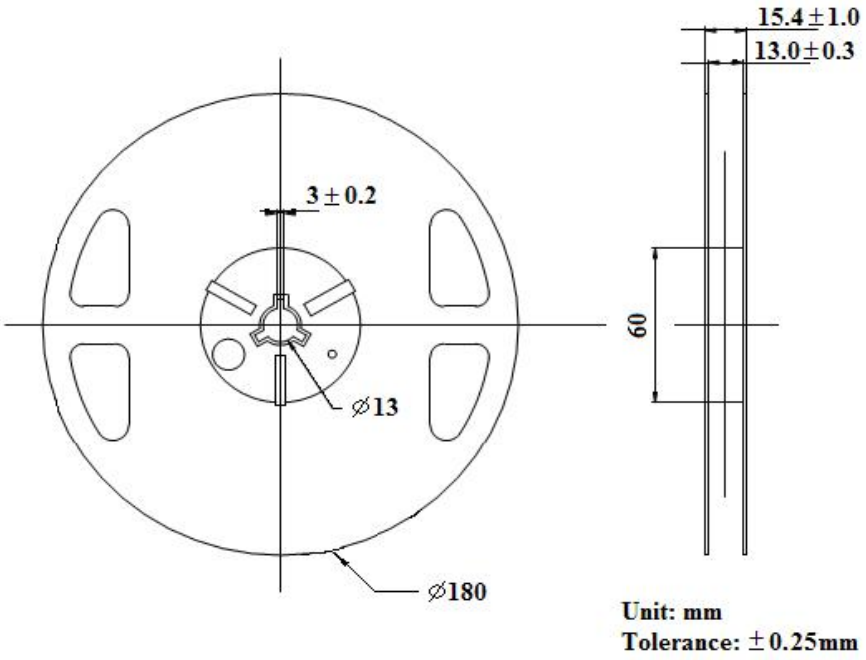






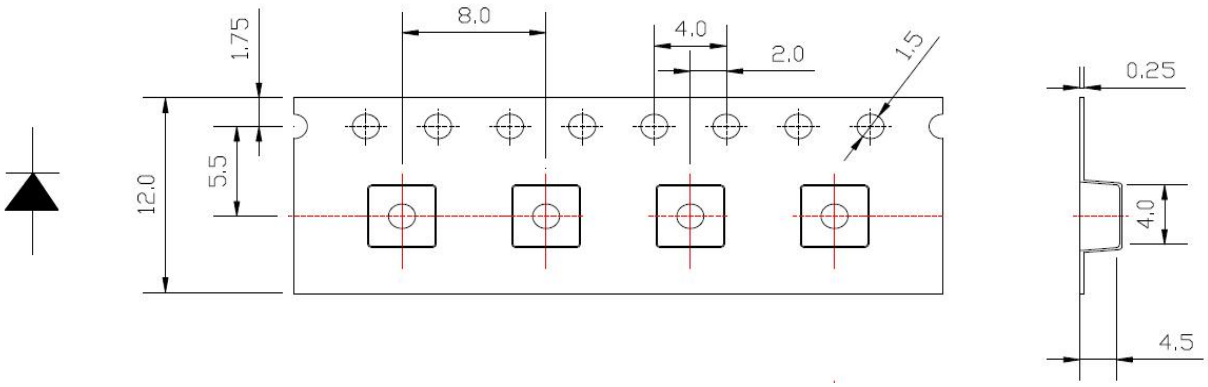


Reel Dimensions



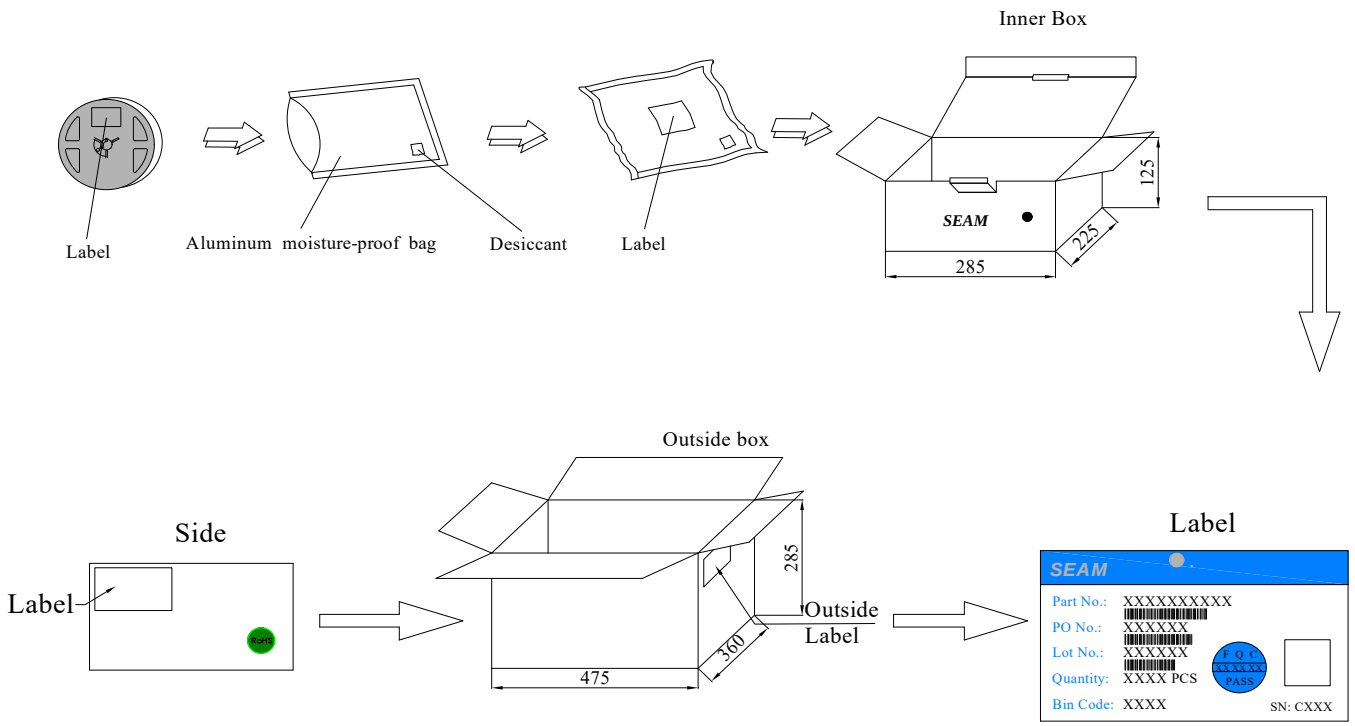
Carrier Tape Dimensions

Loaded quantity 500 pcs per reel.



Packing & Label Specifications

Moisture Resistant Packaging:



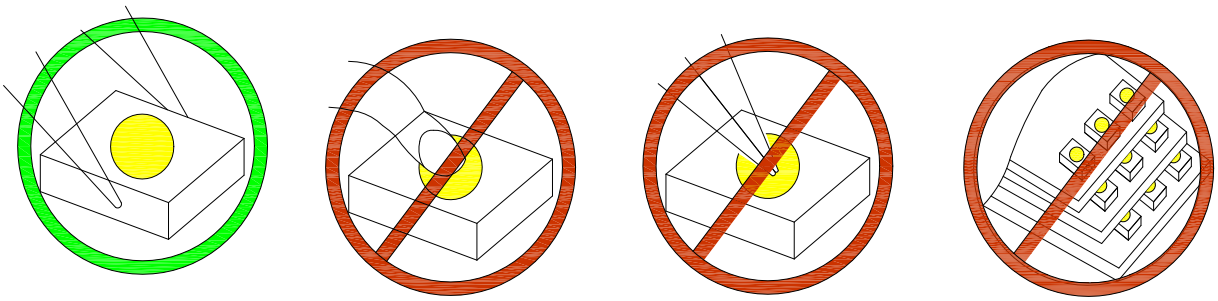
CAUTIONS

1. Handling Precautions:

1.1. Handle the component along the side surfaces by using forceps or appropriate tools.

1.2. Do not directly touch or handle the silicone lens surface. It may damage the internal circuitry.

1.3. Do not stack together assembled PCBs containing exposed LEDs. Impact may scratch the silicone lens or damage the internal circuitry.



Compare to epoxy encapsulant that is hard and brittle, silicone is softer and flexible. Although its characteristic significantly reduces thermal stress, it is more susceptible to damage by external mechanical force. As a result, special handling precautions need to be observed during assembly using silicone encapsulated LED products. Failure to comply might lead to damage and premature failure of the LED.

2. Storage

2.1. Do not open moisture proof bag before the products are ready to use.

2.2. Before opening the package, the LEDs should be kept at 30°C or less and 60%RH or less.

2.3. The LEDs should be used within a year.

2.4. After opening the package, the LEDs should be kept at 30°C or less and 60%RH or less.

2.5. The LEDs should be used within 168 hours after opening the package.

2.6. If the moisture adsorbent material has fabled away or the LEDs have exceeded the storage time, baking treatment should be performed using the following conditions. Baking treatment: 65±5°C for 24 hours

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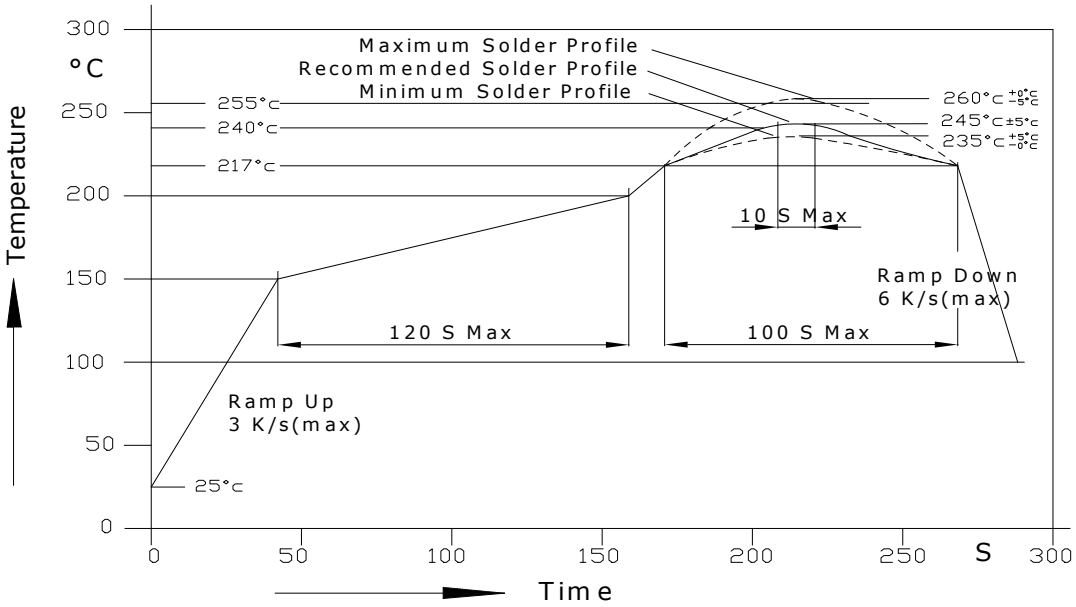
3. **Soldering Condition**

3.1. Pb-free solder temperature profile



3.2. Reflow soldering should not be done more than two times.

3.3. When soldering, do not put stress on the LEDs during heating.

3.4. After soldering, do not warp the circuit board.

3.5. Recommended soldering conditions:

Reflow soldering		Soldering iron	
Pre-heat	150~200°C	Temperature	300°C Max.
Pre-heat time	120 sec. Max.	Soldering time	3 sec. Max.
Peak temperature	260°C Max.		(one time only)
Soldering time	10 sec. Max.(Max. two times)		

3.6. Because different board designs use different number and types of devices, solder pastes, reflow ovens, and circuit boards, no single temperature profile works for all possible combinations.

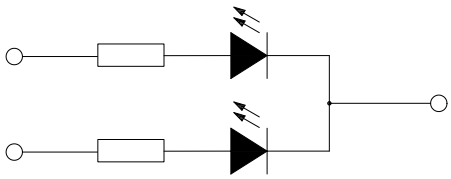
However, you can successfully mount your packages to the PCB by following the proper guidelines and PCB-specific characterization.

4. **Drive Method**

4.1. An LED is a current-operated device. In order to ensure intensity uniformity on multiple LEDs connected in parallel in an application, it is recommended that a current limiting resistor be incorporated in the drive circuit, in series with each LED as shown in Circuit A below.

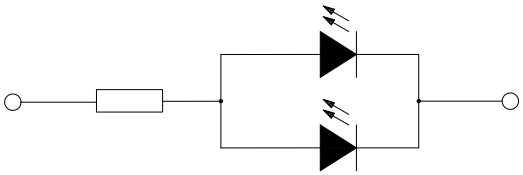
Circuit model A

LED



Circuit model B

LED



a. Recommended circuit.

b. The brightness of each LED might appear different due to the differences in the I-V characteristics of those LEDs.

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